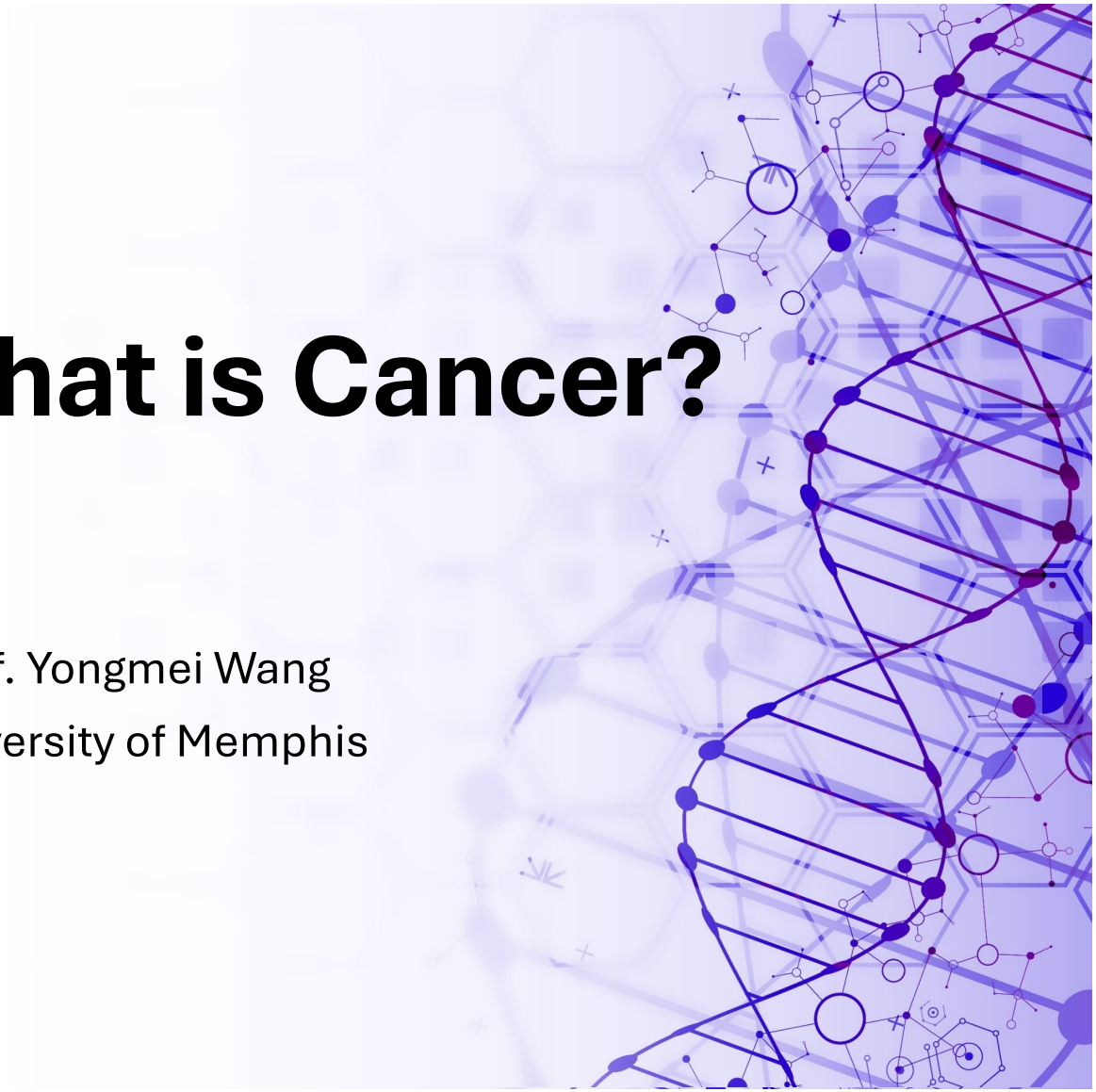
Lecture I: What is Cancer?

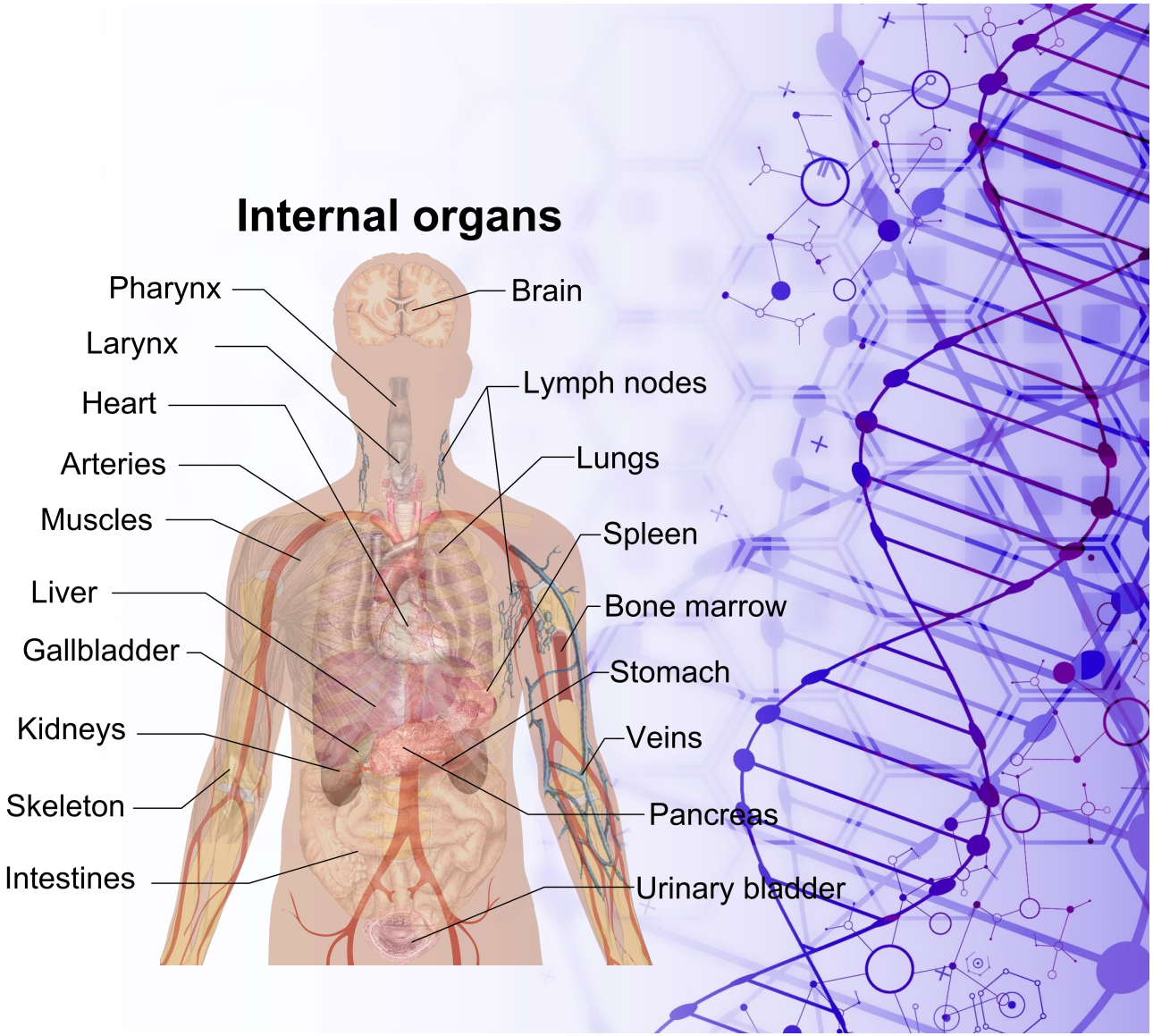


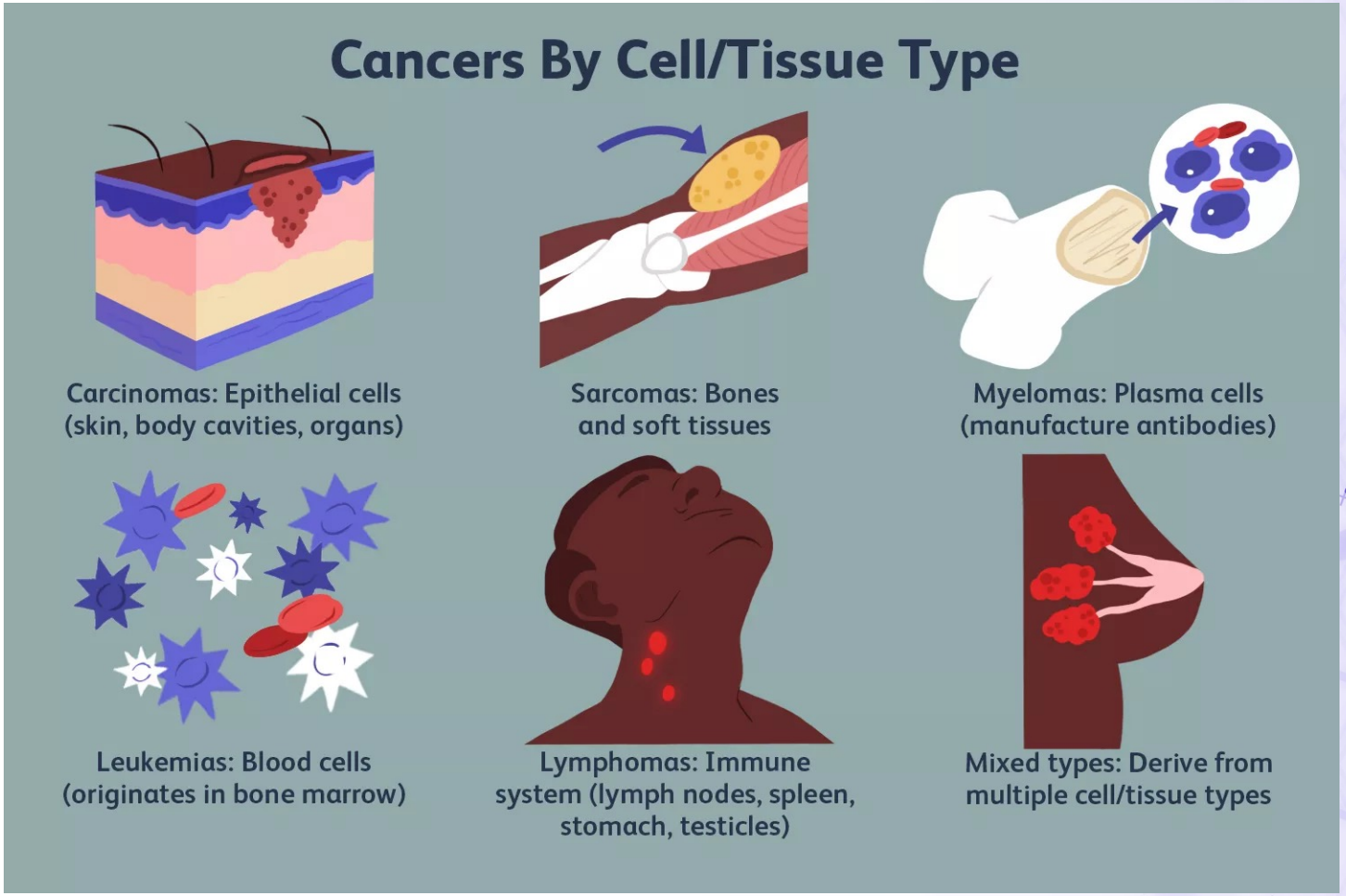
by Prof. Yongmei Wang

The University of Memphis

The human body is made up of trillions of cells, work together to form tissues and organs, like your skin, muscles, and brain. Normally, these cells grow, divide, and die in a very controlled way to keep your body healthy.

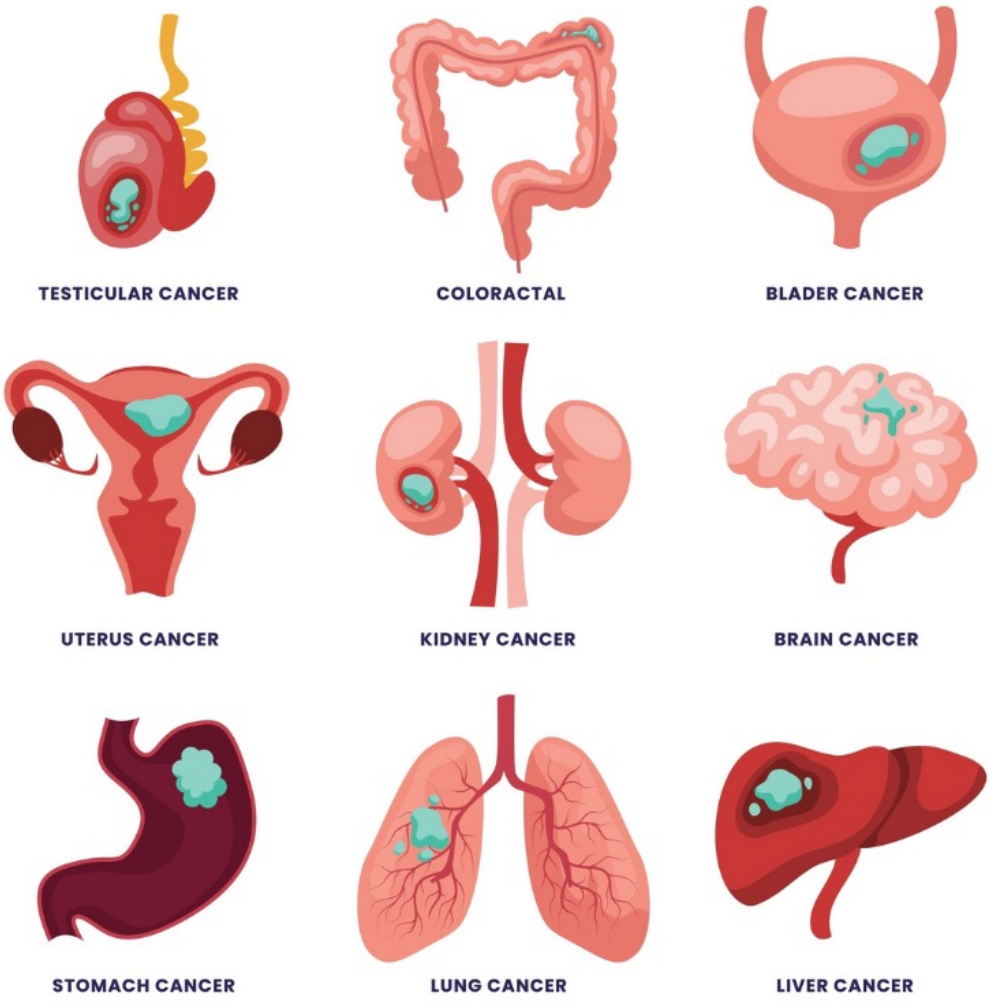
But sometimes, cells stop following the rules. They start growing and dividing out of control, ignoring the signals that usually tell them when to stop. When this happens, a group of these abnormal cells can form a lump called a tumor, or they can spread to other parts of the body. This uncontrolled growth of abnormal cells is what we call **cancer**."

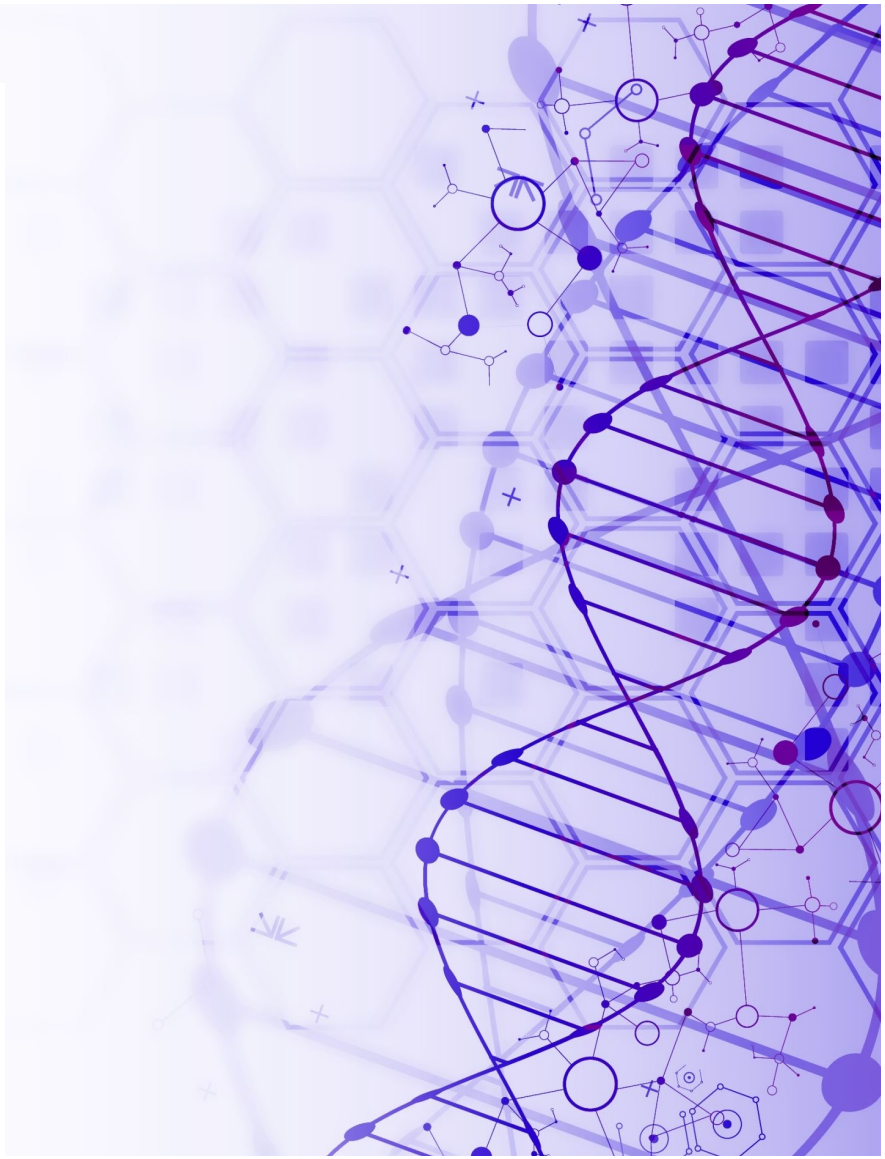




- https://www.verywellhealth.com/more-cancer-types-4158486

Cancer by organ





Estimated New Cases

Male			Female		
Prostate	299,010	29%	Breast	310,720	32%
Lung & bronchus	116,310	11%	Lung & bronchus	118,270	12%
Colon & rectum	81,540	8%	Colon & rectum	71,270	7%
Urinary bladder	63,070	6%	Uterine corpus	67,880	7%
Melanoma of the skin	59,170	6%	Melanoma of the skin	41,470	4%
Kidney & renal pelvis	52,380	5%	Non-Hodgkin lymphoma	36,030	4%
Non-Hodgkin lymphoma	44,590	4%	Pancreas	31,910	3%
Oral cavity & pharynx	41,510	4%	Thyroid	31,520	3%
Leukemia	36,450	4%	Kidney & renal pelvis	29,230	3%
Pancreas	34,530	3%	Leukemia	26,320	3%
All sites	**1,029,080**		**All sites**	**972,060**	

Estimated Deaths

Male			Female		
Lung & bronchus	65,790	20%	Lung & bronchus	59,280	21%
Prostate	35,250	11%	Breast	42,250	15%
Colon & rectum	28,700	9%	Pancreas	24,480	8%
Pancreas	27,270	8%	Colon & rectum	24,310	8%
Liver & intrahepatic bile duct	19,120	6%	Uterine corpus	13,250	5%
Leukemia	13,640	4%	Ovary	12,740	4%
Esophagus	12,880	4%	Liver & intrahepatic bile duct	10,720	4%
Urinary bladder	12,290	4%	Leukemia	10,030	3%
Non-Hodgkin lymphoma	11,780	4%	Non-Hodgkin lymphoma	8,360	3%
Brain & other nervous system	10,690	3%	Brain & other nervous system	8,070	3%
All sites	**322,800**		**All sites**	**288,920**	

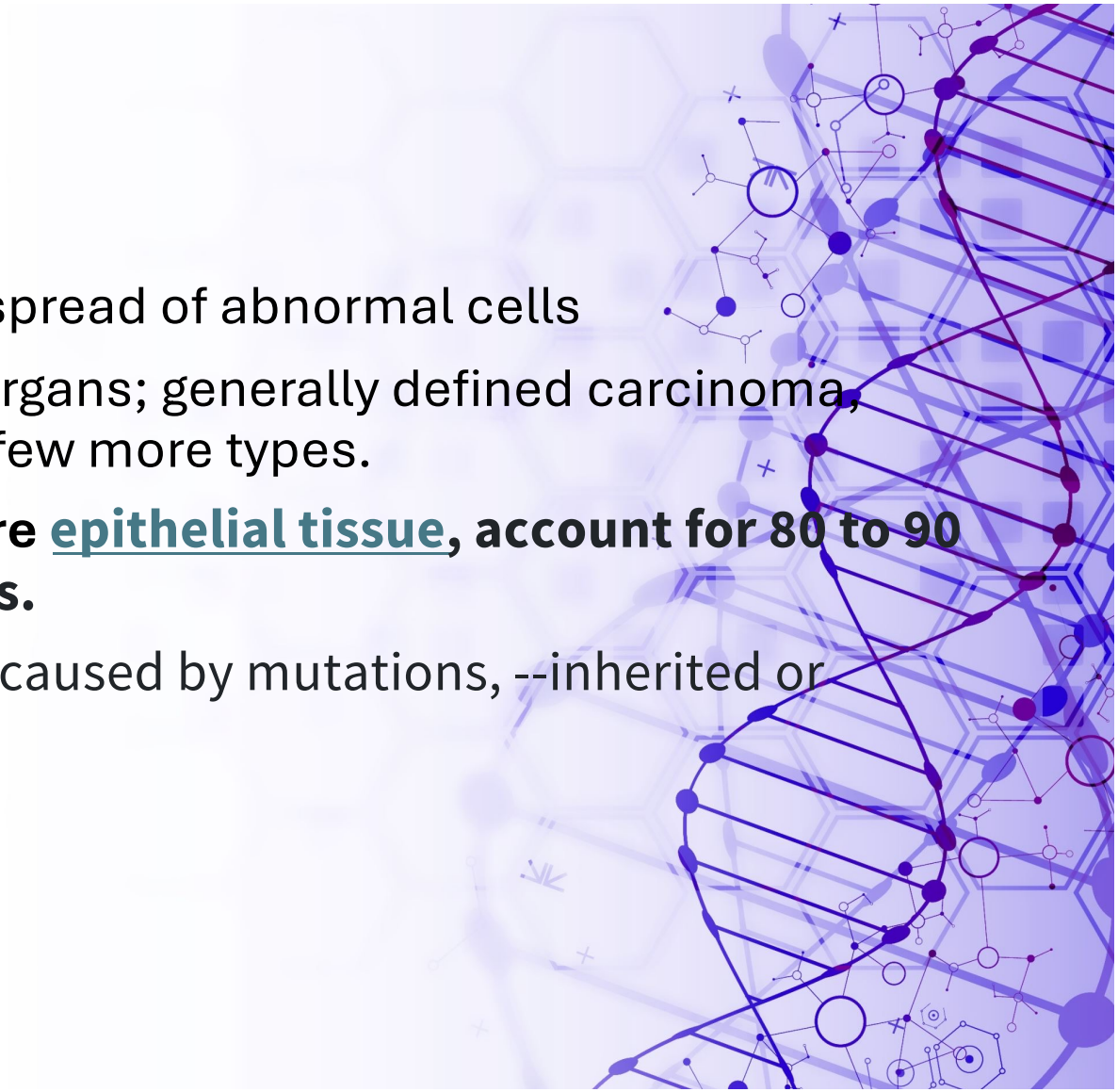
Estimates are rounded to the nearest 10, and cases exclude basal cell and squamous cell skin cancers and in situ carcinoma except urinary bladder. Estimates do not include Puerto Rico or other US territories. Ranking is based on modeled projections and may differ from the most recent observed data.

©2024, American Cancer Society, Inc., Surveillance and Health Equity Science

https://www.cancer.org/research/acs-research-news/facts-and-figures-2024.html

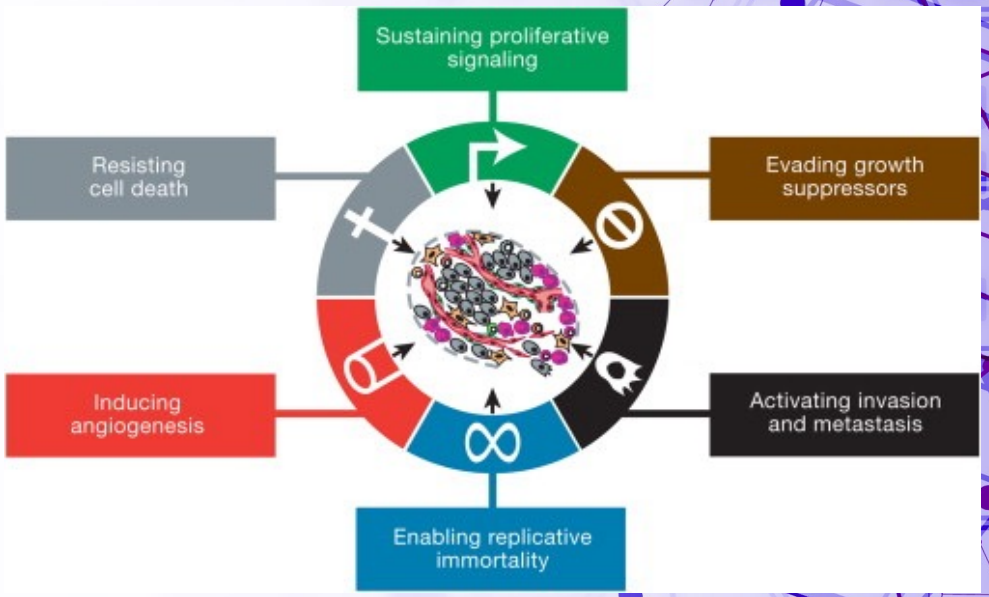
What is cancer?

- Uncontrolled growth and spread of abnormal cells
- Can grow on any tissues/organs; generally defined carcinoma, sarcoma, leukemia and a few more types.
- **Carcinoma cell origins are epithelial tissue, account for 80 to 90 percent of all cancer cases.**
- Cancers are believed to be caused by mutations, --inherited or acquired



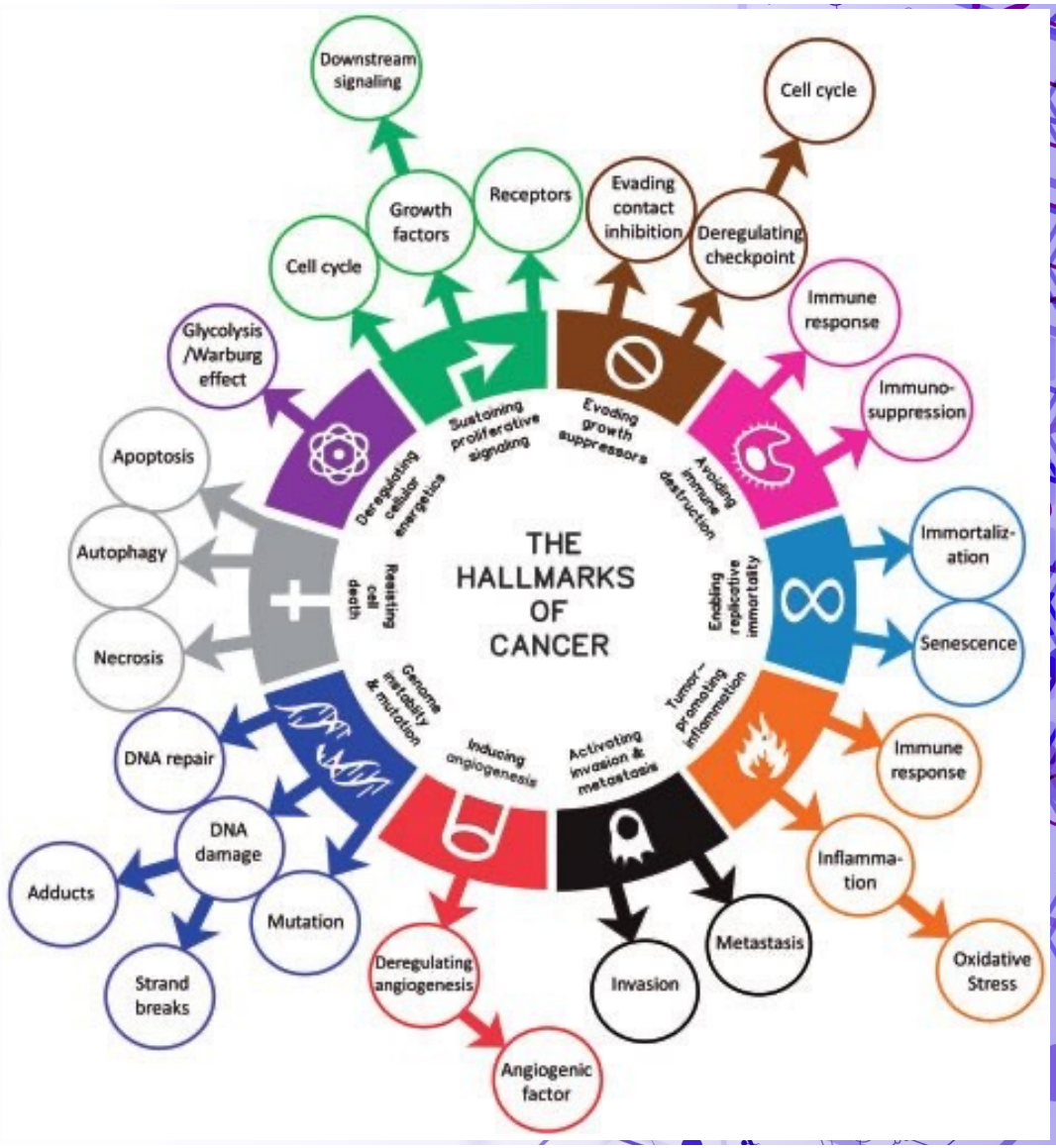
hallmarks of cancers

- **Sustaining proliferative signaling**
- **Evading growth suppressors**
- **Activating invasion and metastasis**
- **Enabling replicative immortality**
- **Inducing angiogenesis**
- **Resisting cell death**



Hallmarks of Cancer: The Next Generation https://www.cell.com/fulltext/S0092-8674(11)00127-9

- **Sustaining proliferative signaling**
- **Evading growth suppressors**
- **Activating invasion and metastasis**
- **Enabling replicative immortality**
- **Inducing angiogenesis**
- **Resisting cell death**
- **Avoiding immune destruction**
- Tumor-Promoting Inflammation
- **Genome Instability and Mutation**
- Deregulating Cellular Energetics



Cancer Hallmarks Analytics Tool (CHAT): a text mining approach to organize and evaluate scientific literature on cancer, https://doi.org/10.1093/bioinformatics/btx454

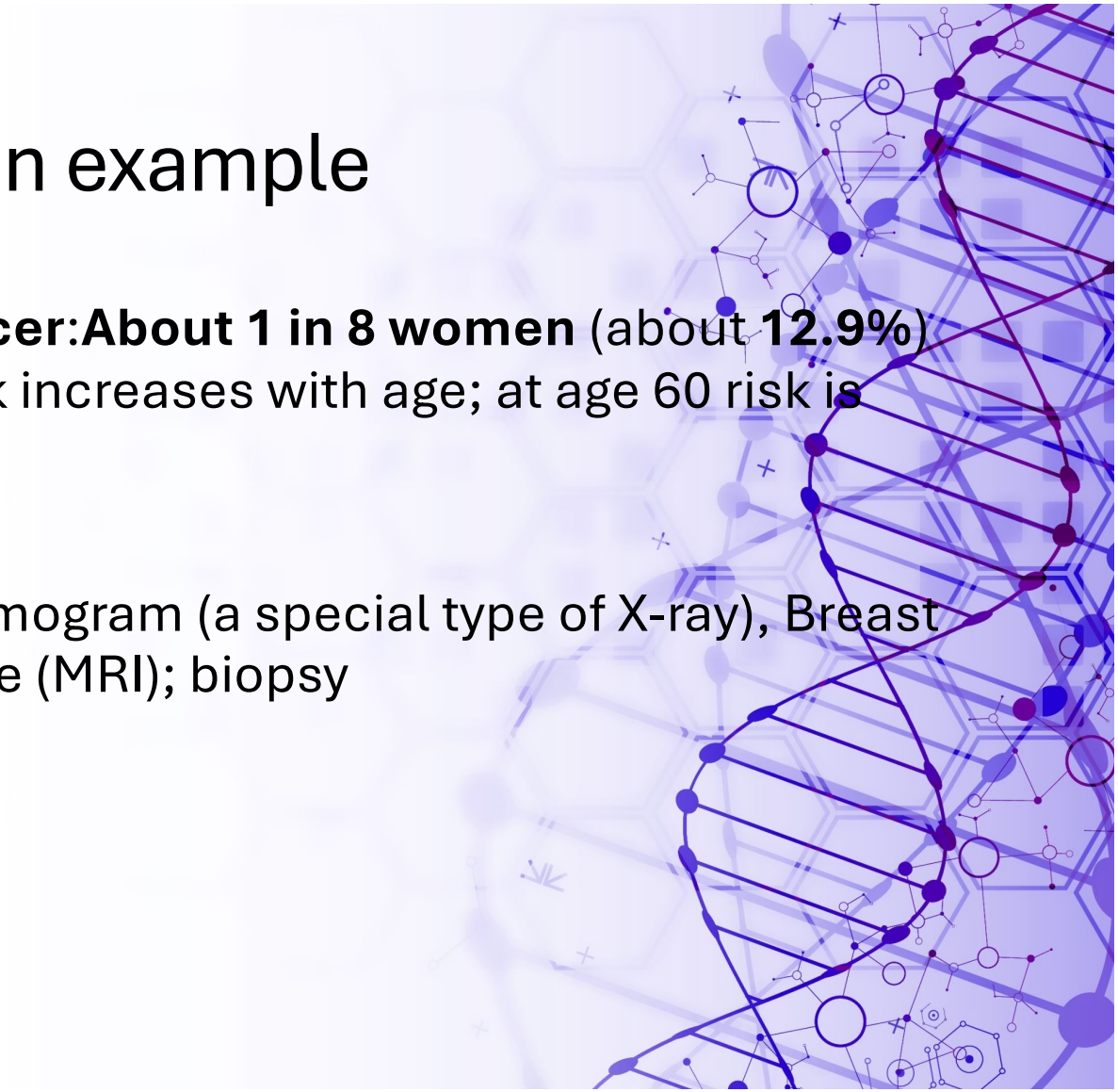
Areas of Cancer Research

- **Mechanistic study**: figuring out what mutations cause cancer, how cancer cells cycle differs from normal cells at molecular and cellular levels
- **Metastasis & Invasion:** Mechanisms of cancer spread and colonization.
- **Early Detection and Diagnosis**: How to detect cancer? Biomarker Discovery & Validation (Blood, Urine, Saliva), Imaging Technologies, Pathology Automation & Digital Pathology
- **Cancer Therapeutics & Drug Development**: Chemotherapy Mechanisms & Resistance, Hormonal Therapies, Immunotherapies , Radiation Therapy Optimization
- **Precision Oncology (Personalized Medicine): matching treatment to individual tumor profiles**
- **Cancer Survivorship & Supportive Care:** Improving life during and after cancer treatment.
- **Clinical Trials & Translational Research**

Breast Cancer as an example

- **Frequency of breast cancer**:**About 1 in 8 women** (about **12.9%**) in lifetime (by age 80); Risk increases with age; at age 60 risk is 3.5%

- **Detection method:** Mammogram (a special type of X-ray), Breast magnetic resonance image (MRI); biopsy



Staging for breast cancers

- Use Stages 0-IV to describe breast cancer
  - **Stage 0**: (pre-cancer): Called **ductal carcinoma in situ (DCIS)**. Abnormal cells are only in the milk ducts and haven't spread. Highly treatable and **not invasive** yet.
  - **Stage I**: Small, Early-Stage Cancer**,** Often found in the breast only, or possibly in a few nearby lymph nodes, usually very treatable
  - **Stage II**: Tumor may be larger or has spread to a few nearby lymph nodes. Still mostly limited to the breast area. Treatment depends on tumor size and lymph node involvement
  - **Stage III**: Cancer has spread to several nearby lymph nodes, possibly nearby tissues like the chest wall or skin. Often called **locally advanced breast cancer**.
  - **Stage IV**: Metastatic Cancer (Advanced Stage), Cancer has spread (metastasized) to **other parts of the body** such as bone, liver, lung and brain;

Breast Cancer Subtypes (Simplified Overview)

Hormone Receptor-Positive (HR+)

Key Feature: Uses hormones to grow (ER/PR positive)
Treatment: Hormone therapy (Tamoxifen, etc.)

HER2-Positive

Key Feature: Extra HER2 protein; fast-growing
Treatment: Targeted therapy (Herceptin, etc.)

Triple-Negative (TNBC)

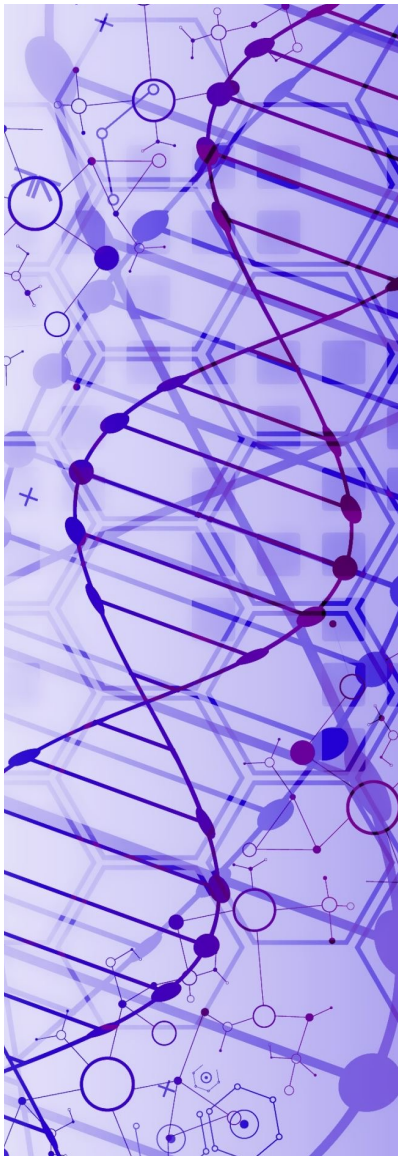
Key Feature: No ER, PR, or HER2; aggressive
Treatment: Chemotherapy, sometimes immunotherapy

Inflammatory Breast Cancer (IBC)

Key Feature: Red, swollen breast; very aggressive
Treatment: Combo of chemo, surgery, radiation

Ductal / Lobular Carcinomas

Key Feature: Based on origin in breast tissue
Treatment: Depends on other features above



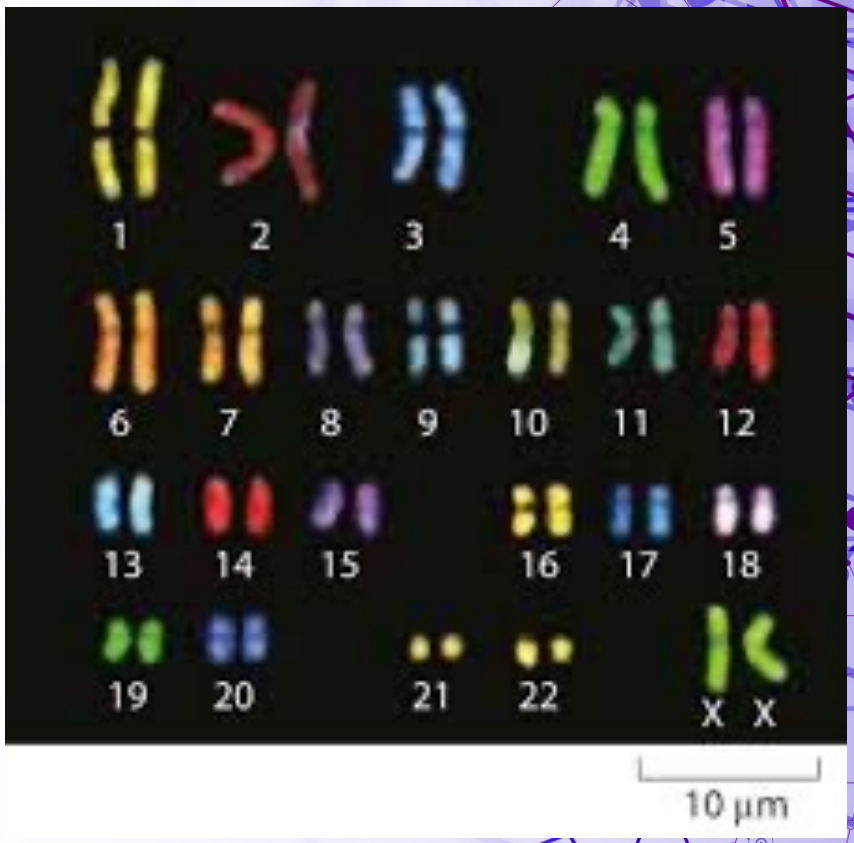
Big data in cancer biology

- It happened after the human genome project completed (~2003).
- The Cancer Genome Atlas Program (TCGA) launched in 2006
- Big data fueled by the low costs of sequencing

https://www.cancer.gov/ccg/research/genome-sequencing/tcga

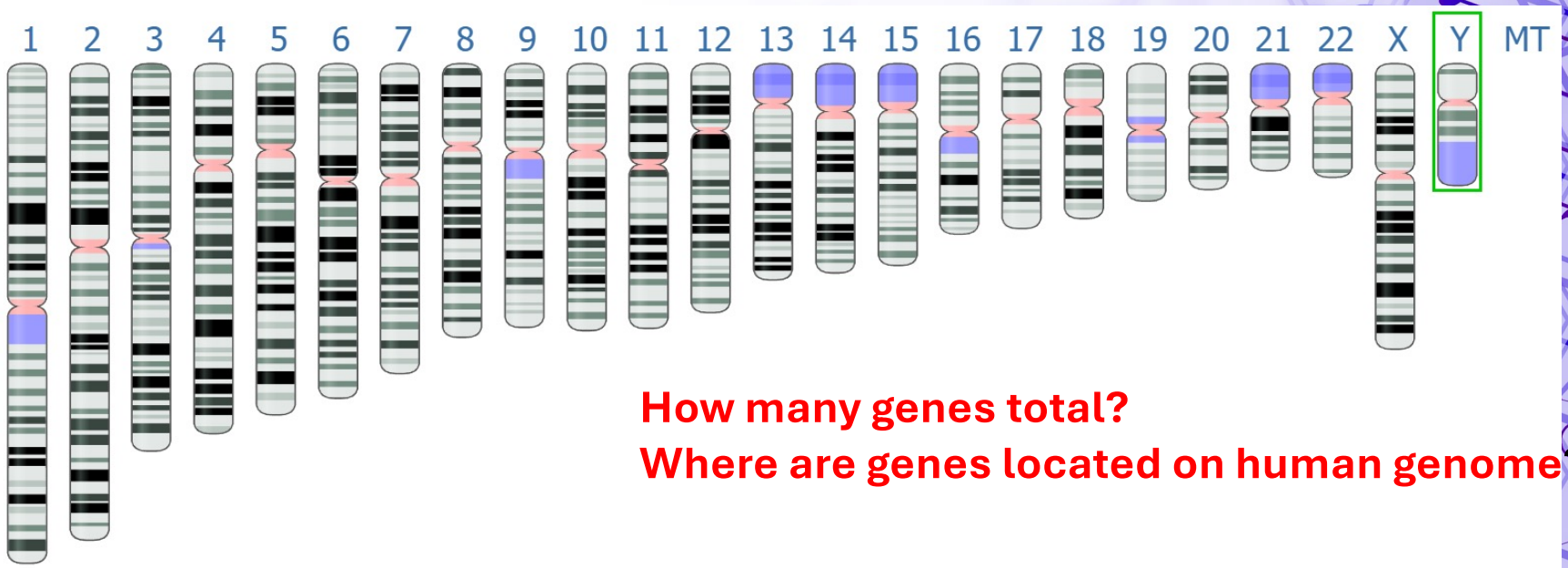
Basic genetics of human

- Human has 23 pairs of chromosome
- Human genome project – sequences all DNA bp in human genome :3 billions base pairs (bps) ($3x10^9$ bps)
- Provide a human genome references



First Human Genome Project

- Started in 1990, completed by 2003;
- Initially planned for 20-year project, $3 billion USD effort
- Used Gene Mapping and DNA sequencing



How many genes total?
Where are genes located on human genome?

Link to Genome data viewer: https://www.ncbi.nlm.nih.gov/genome/gdv/browser/genome/?id=GCF_000001405.40

First Activity

I. Pick a cancer type and do some research to write an article that include answers to the following questions
   - Frequencies of this type of cancer occur among the population
   - How is this type of cancer typically identified/detected? Any known factors that give a person a high risk of getting that specific type of cancer?
   - What stagings are used to describe this cancer? What subtypes are used in this cancer?
   - What are the current available treatments for this cancer?
   - What is the average life expectancy for the patient?

II. Start a personal blog on Github and post your research there

How to start a personal blog at Github?

- First visit my personal blog on Github and find the tutorial I used to create it: https://yongmeiwang.github.io/
- Visit my student personal blog that gives you another tutorial on how to start personal blog: https://kombateldridge.github.io/2022/06/14/GitHub-Page-Tutorial.html